

Smart Agricultural Production Optimization Engine (OptiCrop)

Entity Relationship Diagram Documentation:

1. Introduction

The Entity Relationship (ER) Diagram represents the database structure of the **Smart Agricultural Production Optimization Engine (OptiCrop)**. It illustrates the major entities, their attributes, and the relationships between them. The ER diagram provides a logical view of how agricultural data is stored and managed for crop prediction and recommendation.

2. Entities Involved

The system contains seven main entities:

- User
- SoilData
- Crop
- Dataset
- MLModel
- Prediction
- Report

3. Primary Keys

Entity	Primary Key
User	user_id
SoilData	soil_id
Crop	crop_id
Dataset	dataset_id
MLModel	model_id
Prediction	prediction_id
Report	report_id

4. Relationships

- One **User** can submit multiple **SoilData** records (1:N).
- One **SoilData** record generates one **Prediction** (1:1).
- One **Crop** can appear in multiple **Prediction** records (1:N).

- One **MLModel** generates multiple **Prediction** records (1:N).
 - One **Dataset** is used to train multiple **MLModel** records (1:N).
 - One **Prediction** can generate multiple **Reports** (1:N).
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5. Foreign Keys

Entity	Foreign Key
SoilData	user_id
Prediction	soil_id
Prediction	crop_id
Prediction	model_id
MLModel	dataset_id
Report	prediction_id

6. Cardinality

- One user can submit many soil records.
 - Each soil record produces one prediction.
 - A crop may be recommended in many predictions.
 - One ML model can serve many predictions.
 - One dataset trains multiple models.
 - One prediction can generate one reports containing the summary and recomendations
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7. Normalization

The database design follows normalization principles up to Third Normal Form (3NF), reducing redundancy, improving data integrity, and ensuring efficient storage and maintenance.

Information is divided into separate entities:

- User Information
- Soil Data
- Crop Information
- Machine Learning Models
- Dataset Details
- Prediction Results
- Report Generation

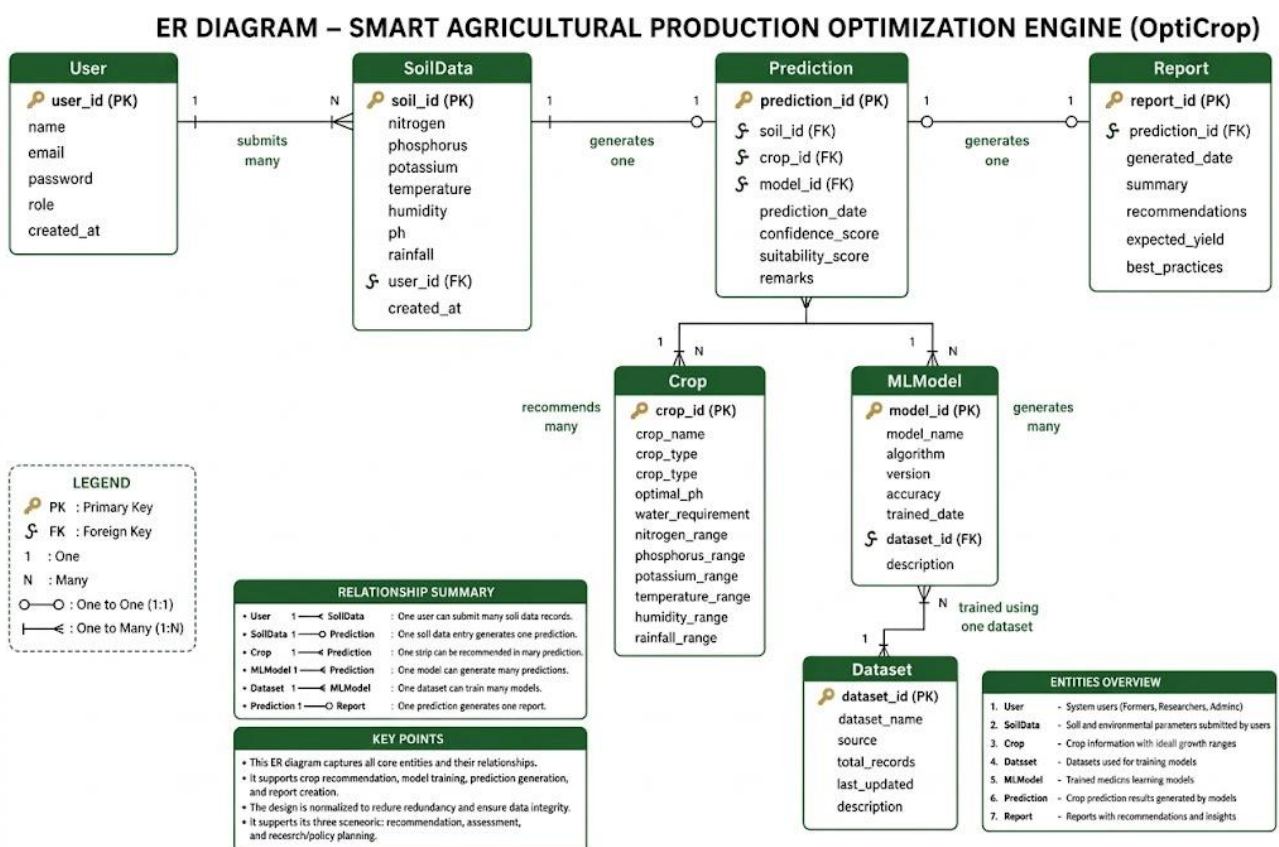
This structure improves data consistency and simplifies maintenance.

8. Use Case Coverage

The ER model supports:

- User registration
- Soil data collection
- Crop prediction
- ML model management
- Dataset management
- Report generation
- Agricultural recommendations

9. ER Diagram



10. Conclusion

The ER diagram provides a scalable and well-structured database design for the OptiCrop system. It supports efficient storage of user, soil, crop, prediction, and report data while maintaining data integrity through well-defined relationships and normalization.